Task

SIEM

(Theoretical Knowledge, Practical Application)

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1. SOC Fundamentals and Operations

1.1 Introduction to Security Operations Center (SOC)

A Security Operations Center (SOC) is a centralized unit responsible for monitoring, detecting, analyzing, and responding to cybersecurity threats within an organization. The primary goal of a SOC is to ensure the security of an organization's information systems by continuously monitoring network activity, identifying potential threats, and responding quickly to security incidents.

SOC teams operate 24/7 to detect malicious activities and prevent cyber attacks before they cause damage. They rely on various security tools such as Security Information and Event Management (SIEM), Intrusion Detection Systems (IDS), Intrusion Prevention Systems (IPS), and Endpoint Detection and Response (EDR) tools to collect and analyze security data from multiple sources.

The SOC plays an important role in maintaining the security posture of an organization by performing proactive threat detection, incident response, and continuous monitoring of systems and networks.

1.2 Purpose of a SOC

The primary purpose of a Security Operations Center is to protect an organization's infrastructure, data, and users from cyber threats. SOC teams continuously monitor networks, endpoints, servers, and applications to detect suspicious activities.

The main objectives of a SOC include:

- **Proactive Threat Detection:** Identifying potential security threats before they impact the organization.
- **Incident Response:** Investigating and responding to security incidents quickly and effectively.
- **Continuous Monitoring:** Monitoring network traffic, logs, and system activity in real time.
- **Threat Intelligence Integration:** Using external threat intelligence sources to detect known attack patterns.

By performing these functions, SOC teams help reduce the risk of data breaches, system compromises, and other cybersecurity incidents.

1.3 Roles in a SOC Team

A Security Operations Center consists of different levels of analysts who handle security incidents at different stages of investigation.

Tier 1 Analyst (SOC Analyst)

Tier 1 analysts are responsible for monitoring security alerts generated by SIEM tools and other monitoring systems. Their main responsibility is to perform the initial analysis of alerts and determine whether they represent a real security threat.

Responsibilities include:

- Monitoring security alerts
- Performing basic log analysis
- Identifying suspicious activity
- Escalating incidents to higher-level analysts when necessary

Tier 2 Analyst (Incident Responder)

Tier 2 analysts investigate alerts escalated by Tier 1 analysts. They perform deeper analysis to understand the cause of the security incident and determine whether the system has been compromised.

Responsibilities include:

- Investigating suspicious activities
- Correlating logs from multiple systems
- Identifying Indicators of Compromise (IOCs)
- Conducting deeper forensic analysis

Tier 3 Analyst (Threat Hunter)

Tier 3 analysts are senior security professionals responsible for advanced threat hunting and attack analysis. They look for hidden threats that may not trigger automatic alerts.

Responsibilities include:

- Performing advanced threat hunting
- Investigating complex attacks
- Malware analysis
- Improving detection rules and security policies

SOC Manager

The SOC Manager oversees the entire SOC team and ensures that security operations run efficiently. They are responsible for managing incident response processes and coordinating with other teams within the organization.

Responsibilities include:

- Managing SOC operations
- Developing security policies
- Coordinating incident response
- Improving security monitoring strategies

1.4 Key Functions of a SOC

SOC teams perform several critical security functions to protect organizational infrastructure.

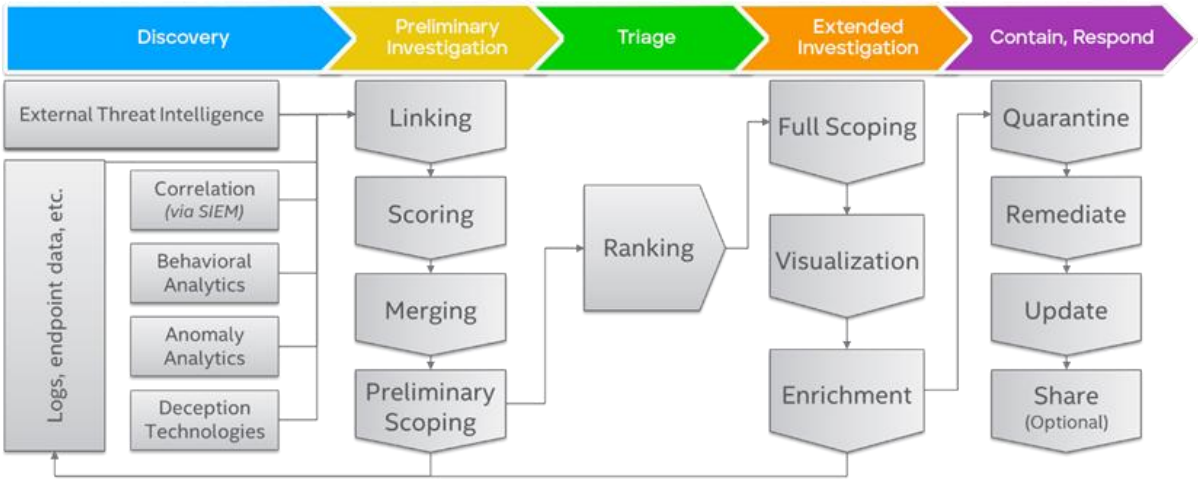

Figure: Security Operations Center Workflow

Log Analysis

Log analysis involves collecting and examining logs from various systems such as servers, firewalls, applications, and endpoints. By analyzing these logs, SOC analysts can detect unusual behavior or suspicious activities that may indicate a security threat.

Alert Triage

Alert triage involves prioritizing security alerts based on their severity and potential impact. SOC analysts evaluate alerts generated by monitoring tools and determine whether they require further investigation.

Alerts are typically classified into different severity levels such as low, medium, high, and critical.

Threat Intelligence Integration

Threat intelligence provides information about known threats, malicious IP addresses, domains, and attack techniques. SOC teams integrate threat intelligence feeds into their monitoring systems to identify known attack patterns and improve threat detection.

Threat intelligence platforms help SOC teams stay updated with the latest cyber threats and vulnerabilities.

1.5 SOC Frameworks

SOC operations are often guided by well-known cybersecurity frameworks that help organizations improve their security practices.

NIST Cybersecurity Framework

The NIST Cybersecurity Framework provides guidelines for managing and reducing cybersecurity risks. It consists of five main functions:

- Identify
- Protect
- Detect
- Respond
- Recover

These functions help organizations create a structured approach to cybersecurity risk management.

MITRE ATT&CK Framework

The MITRE ATT&CK framework is a globally recognized knowledge base that documents real-world cyber attack techniques used by adversaries. It categorizes attacks into tactics and techniques that help security analysts understand how attackers operate.

SOC teams use the MITRE ATT&CK framework to map detected threats to known attack techniques and improve detection and response strategies.

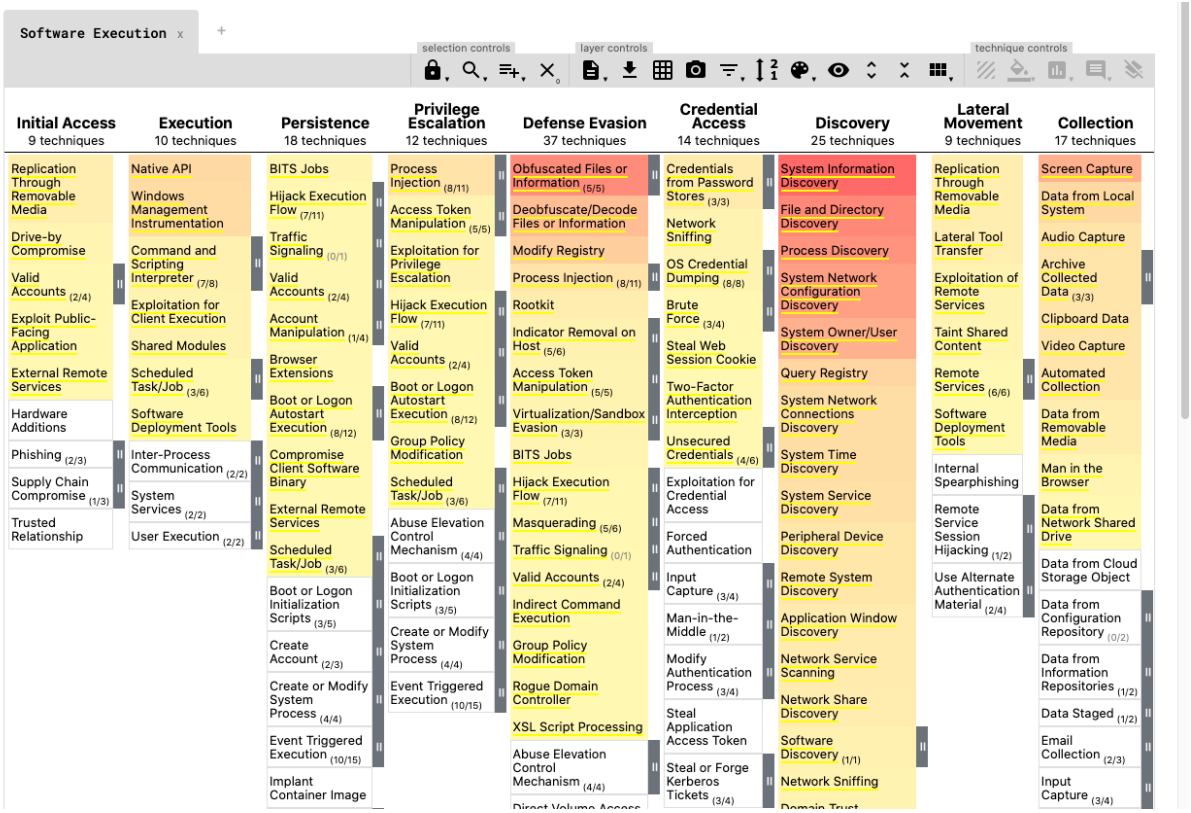

Figure: MITRE ATT&CK Framework for Threat Detection

2. Security Monitoring Basics

2.1 Introduction

Security monitoring is the process of continuously observing and analyzing system and network activities to detect potential security threats. It helps organizations identify anomalies, unauthorized access attempts, and policy violations in real time. Security monitoring plays a crucial role in cybersecurity operations because it allows security teams to detect suspicious activities before they escalate into major security incidents.

Organizations rely on monitoring tools to collect and analyze logs from different sources such as servers, endpoints, network devices, and applications. By analyzing these logs, security analysts can detect abnormal behavior, unauthorized access attempts, or malicious network activity.

2.2 Objectives of Security Monitoring

The primary objective of security monitoring is to identify and respond to security threats as quickly as possible. It enables organizations to maintain visibility into their systems and detect suspicious activities.

Some key objectives include:

- **Detect anomalies:** Identifying unusual system behavior that may indicate a cyber attack.
- **Detect unauthorized access:** Monitoring login attempts and access activities.
- **Identify policy violations:** Detecting activities that violate security policies.
- **Provide real-time alerts:** Generating alerts when suspicious activity is detected.

Effective monitoring ensures that threats are detected early and mitigated before causing damage to the system or network.

2.3 Tools Used in Security Monitoring

Security monitoring relies on several tools that collect and analyze security data.

SIEM (Security Information and Event Management)

SIEM systems collect and analyze logs from multiple sources to detect security threats. These systems correlate data from different systems and generate alerts when suspicious activity is detected.

Examples of SIEM tools include:

- Splunk
- Elastic SIEM
- Wazuh
- IBM QRadar

These tools allow security analysts to visualize security data, investigate alerts, and detect potential attacks.

Network Traffic Analyzers

Network traffic analyzers help security teams monitor network communication and detect malicious traffic patterns.

One commonly used tool is **Wireshark**, which captures and analyzes packets transmitted across a network. By inspecting packet data, analysts can identify suspicious activities such as abnormal connections, unusual protocols, or data exfiltration attempts.

2.4 Key Security Monitoring Metrics

Security monitoring also involves measuring the effectiveness of detection systems using specific metrics.

False Positives

A false positive occurs when a security system incorrectly identifies normal activity as malicious. Excessive false positives can overwhelm security analysts and reduce the efficiency of monitoring systems.

False Negatives

A false negative occurs when a real security threat is not detected by the monitoring system. This is dangerous because the attack may go unnoticed.

Mean Time to Detect (MTTD)

Mean Time to Detect refers to the average time required to detect a security incident after it occurs. A lower MTTD indicates that the monitoring system is effective in identifying threats quickly.

2.5 Practical Monitoring Example

In a practical security monitoring environment, analysts can use tools such as Wireshark to capture network traffic and analyze packet activity. By examining packet details such as source IP address, destination IP address, and protocols used, analysts can identify suspicious behavior.

For example, repeated connection attempts from the same IP address may indicate a brute-force attack. Similarly, unusual network traffic patterns may suggest malware communication or data exfiltration.

Security monitoring tools help analysts investigate these activities and determine whether they represent legitimate traffic or potential threats.

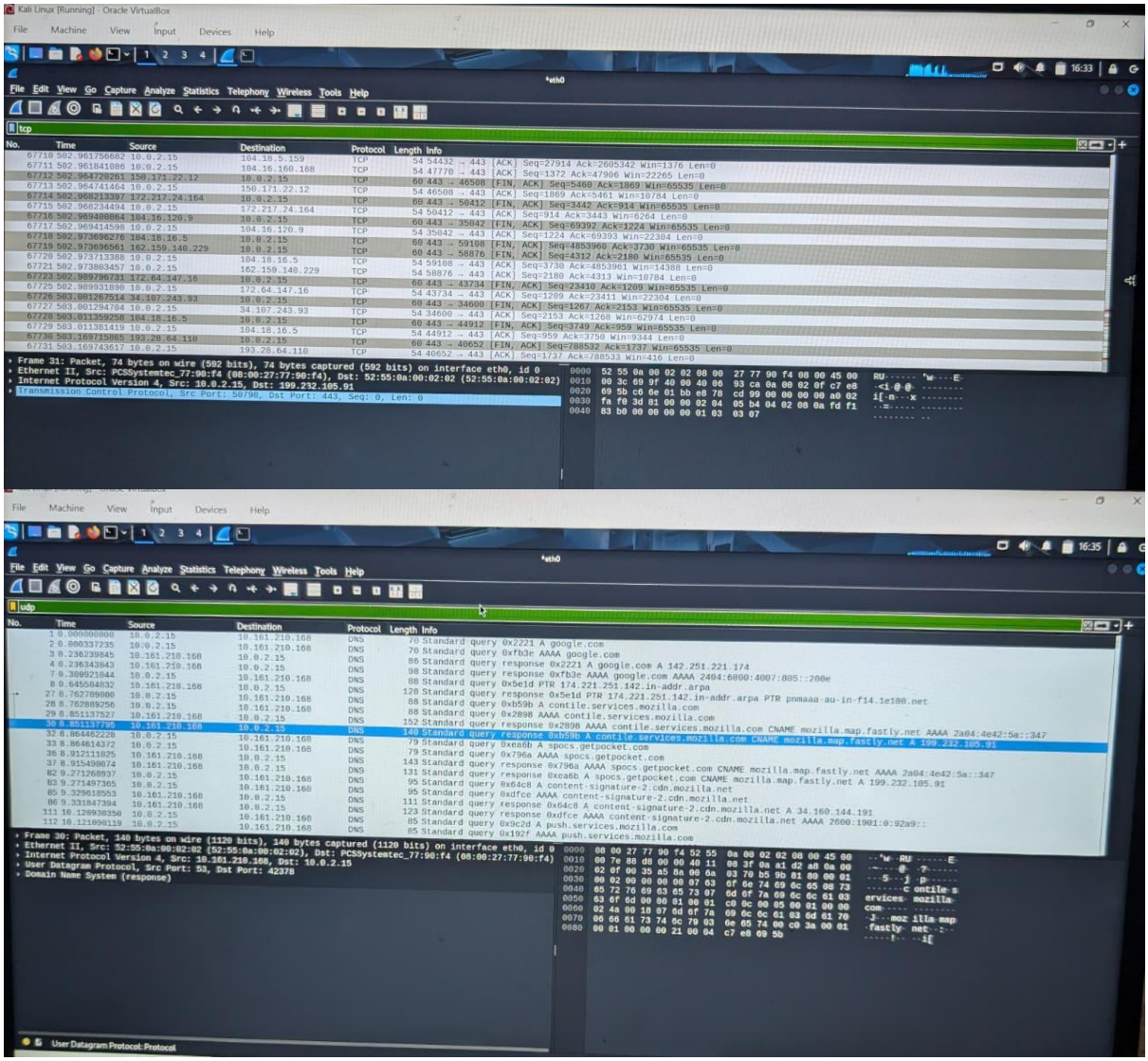

Figure: Network Traffic Monitoring Using Wireshark

3. Log Management Fundamentals

3.1 Introduction

Log management is a critical component of cybersecurity monitoring and incident investigation. Logs provide detailed records of system activities, user actions, network traffic, and application behavior. By collecting and analyzing logs, security analysts can identify suspicious activity, detect security incidents, and investigate potential threats.

Effective log management allows organizations to maintain visibility across their infrastructure and respond quickly to security events. Logs are collected from multiple sources such as operating systems, servers, applications, firewalls, and network devices.

3.2 Log Lifecycle

The log lifecycle consists of several stages that ensure logs are properly collected, processed, stored, and analyzed.

Log Collection

Log collection involves gathering logs from different systems such as servers, applications, and network devices. Tools like **Fluentd, Logstash, and Syslog servers** are commonly used to collect logs from multiple sources.

Log Normalization

Log normalization converts logs from different formats into a standardized format such as **JSON or Common Event Format (CEF)**. This process ensures that logs from various systems can be analyzed efficiently using SIEM tools.

Log Storage

Once logs are collected and normalized, they are stored in centralized storage systems such as SIEM platforms or log management databases. Centralized storage allows analysts to search and analyze logs easily.

Log Retention

Log retention refers to the duration logs are stored before deletion. Organizations often retain logs for several months or years depending on compliance requirements and security policies.

Log Analysis

Log analysis is the process of examining logs to identify security events, anomalies, or malicious activities. Security analysts use log analysis tools and queries to detect suspicious patterns such as repeated login failures, unauthorized access attempts, or abnormal network activity.

3.3 Common Log Types

Several types of logs are used in security monitoring.

Windows Event Logs

Windows Event Logs record system activities such as user logins, application errors, and security events. These logs are important for detecting unauthorized access attempts and suspicious system behavior.

Example event IDs include:

- **Event ID 4625** – Failed login attempts
- **Event ID 4624** – Successful login
- **Event ID 7045** – New service installation

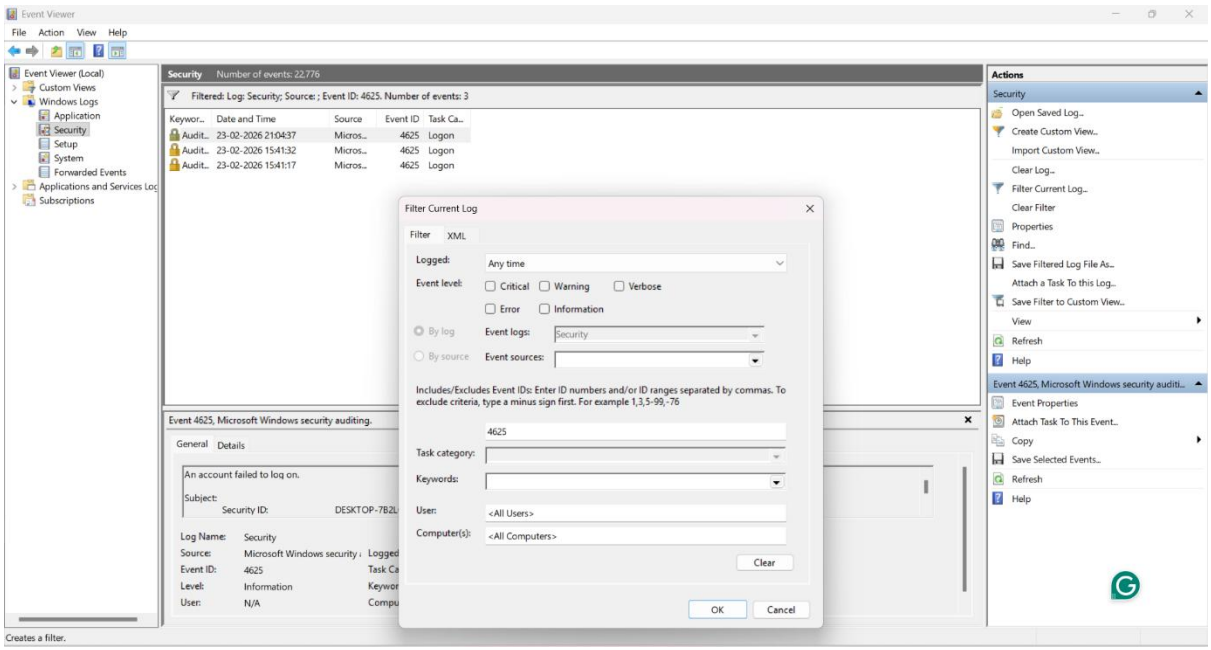

Figure: Windows Event Log Showing Failed Login Attempts (Event ID 4625)

Syslog Logs

Syslog is a standard logging protocol used by Linux systems, network devices, and servers to send log messages to a central logging server.

Syslog logs often contain information about:

- System processes
- Authentication events
- Network activity

HTTP Server Logs

HTTP server logs record web server activity, including requests made by users, IP addresses, and accessed resources. These logs are useful for detecting suspicious web traffic and potential attacks such as SQL injection or brute-force attempts.

3.4 Log Collection Using Fluentd

Fluentd is an open-source data collector used to gather logs from various sources and forward them to centralized log management systems such as Elastic SIEM.

In a lab environment, Fluentd can be configured on an Ubuntu system to collect **Syslog messages**. Test logs can be generated using the following command:

```
logger "Test message"
```

This command generates a test log entry which can then be forwarded to a log management system such as Elastic SIEM for analysis.

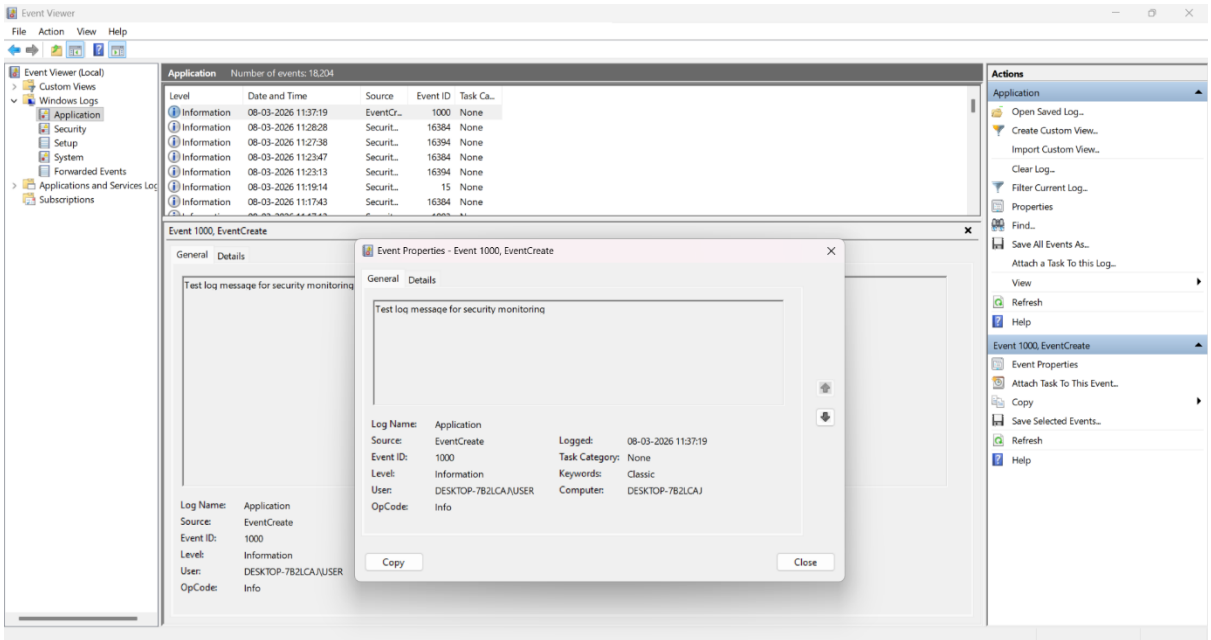

Figure: Generated Test Log Entry Using EventCreate Command

3.5 Log Query Example

Security analysts often use queries to search logs and detect suspicious activities.

For example, in Elastic SIEM a query can be used to detect failed login attempts:

```
source = "security-login-*" EventID = 4625 | stats count by SourceIP
```

This query searches for logs related to failed login attempts and groups them based on the source IP address. Such queries help analysts identify brute-force attacks or unauthorized login attempts.

3.6 Log Normalization Using Logstash

Logstash is a log processing tool used to collect, process, and transform logs into standardized formats.

In a typical setup, Logstash can convert Apache server logs into **JSON format**. This allows logs to be easily analyzed using SIEM tools and log analytics platforms.

Example workflow:

1. Collect Apache access logs
2. Process logs using Logstash
3. Convert logs into JSON format
4. Store the processed logs for analysis

Log normalization improves the efficiency of log analysis by ensuring consistent data formats across different log sources.

4. Security Tools Overview

Security operations rely on various tools that help detect threats, monitor systems, and identify vulnerabilities. These tools enable security analysts to analyze system activities, detect suspicious behavior, and respond to security incidents effectively. Common tools used in security environments include Security Information and Event Management (SIEM) platforms such as Splunk and IBM QRadar, Endpoint Detection and Response (EDR) tools such as CrowdStrike, Intrusion Detection Systems (IDS) like Snort, and vulnerability scanners such as Nessus.

SIEM platforms collect and analyze logs from multiple systems to identify anomalies and potential threats. EDR solutions monitor endpoint activities to detect malicious behavior on devices such as laptops, servers, and workstations. IDS tools inspect network traffic and detect

suspicious patterns that may indicate cyberattacks. Vulnerability scanners analyze systems to identify known security weaknesses that attackers could exploit.

Several tools are available for educational and testing purposes. For example, Splunk Free and Wazuh provide platforms for learning log monitoring and security analytics. Snort can be configured in a sandbox environment to test custom intrusion detection rules. Nessus Essentials allows users to perform vulnerability scans on virtual machines. Additionally, Osquery enables endpoint monitoring through SQL-like queries that analyze system activity.

4.1 Snort Rule Testing

Snort is an open-source intrusion detection system used to monitor network traffic and detect malicious activities based on predefined rules. In this task, Snort was installed on an Ubuntu Server virtual machine to analyze network traffic and detect suspicious behavior.

A custom rule was created in the Snort **local.rules** configuration file to detect HTTP requests containing a malicious domain.

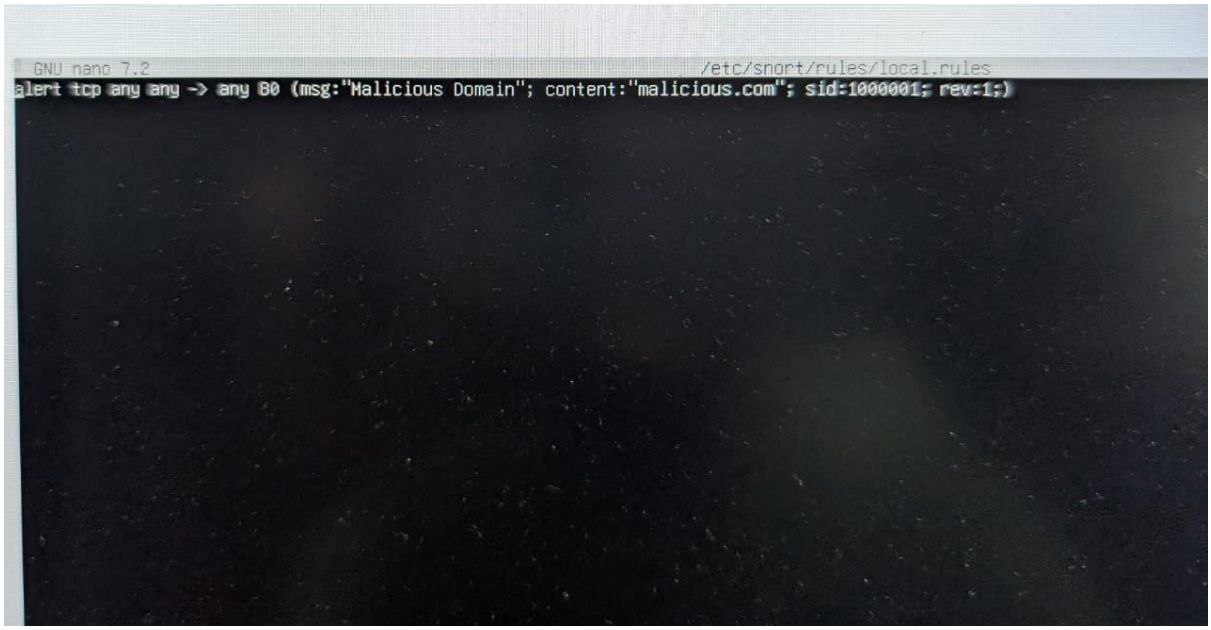

Figure: Custom Snort rule created in /etc/snort/rules/local.rules to detect HTTP traffic containing the domain *malicious.com*.

The rule used in this experiment was:

```
alert tcp any any -> any 80 (msg:"Malicious Domain"; content:"malicious.com"; sid:1000001; rev:1;)
```

This rule instructs Snort to generate an alert whenever TCP traffic directed to port 80 (HTTP) contains the string **“malicious.com”** in the request.

After creating the rule, Snort was executed in console mode on the Ubuntu server to monitor network traffic.

Figure: Snort running in console mode and monitoring network traffic using the custom rule.

To simulate malicious traffic, an HTTP request containing the malicious domain was generated from a Kali Linux machine using the **curl** command.

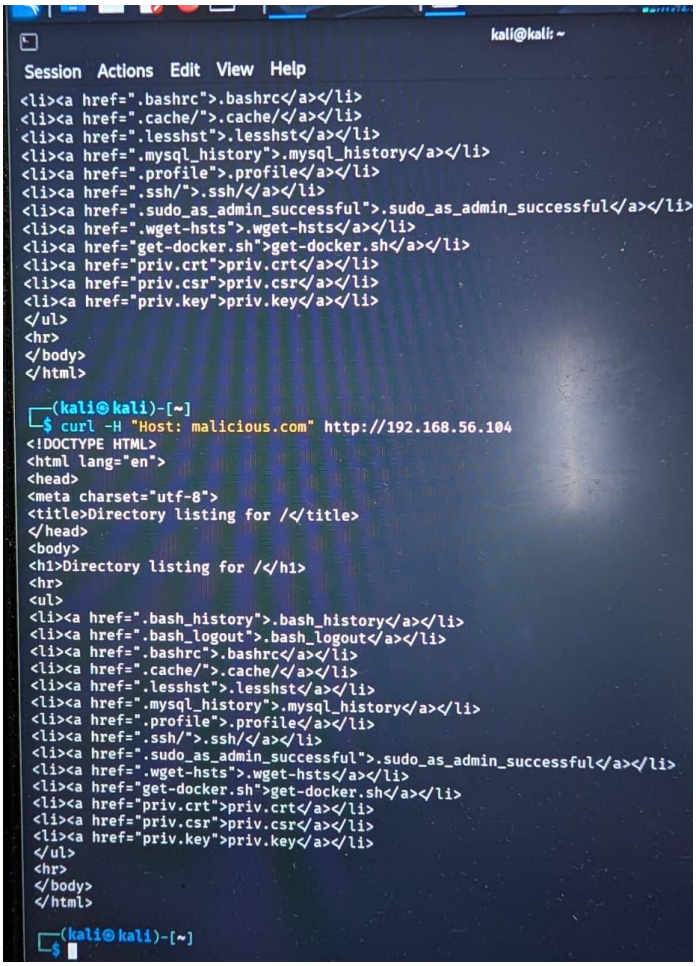

Figure: Kali Linux generating HTTP traffic containing the malicious domain using the curl command.

On the Ubuntu server, a simple HTTP server was started using Python to receive the request and generate network traffic.

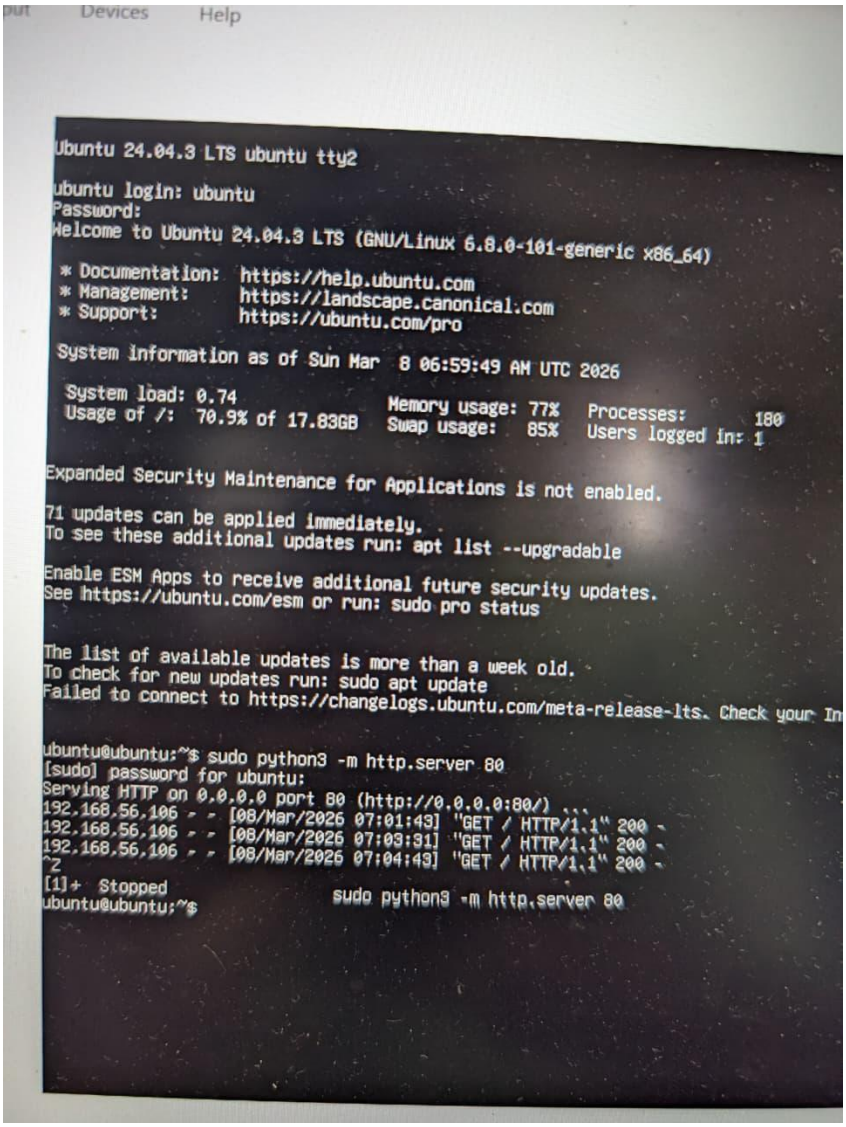

Figure**:** Python HTTP server running on the Ubuntu server to simulate a web service.

When the HTTP request containing the malicious domain was sent, Snort detected the traffic and generated an alert in the console output. This confirmed that the custom Snort rule was functioning correctly and successfully detecting the simulated malicious activity.

4.2 Nessus Vulnerability Scan

Nessus is a widely used vulnerability scanning tool that identifies security weaknesses in systems, applications, and network services. In this task, Nessus Essentials was used to perform a vulnerability scan against the **Metasploitable2** virtual machine, which is intentionally designed with vulnerabilities for testing purposes.

After installing Nessus and accessing the web interface, a **Basic Network Scan** was configured by specifying the IP address of the Metasploitable2 virtual machine. The scan analyzed the system for known vulnerabilities, exposed services, and potential security misconfigurations.

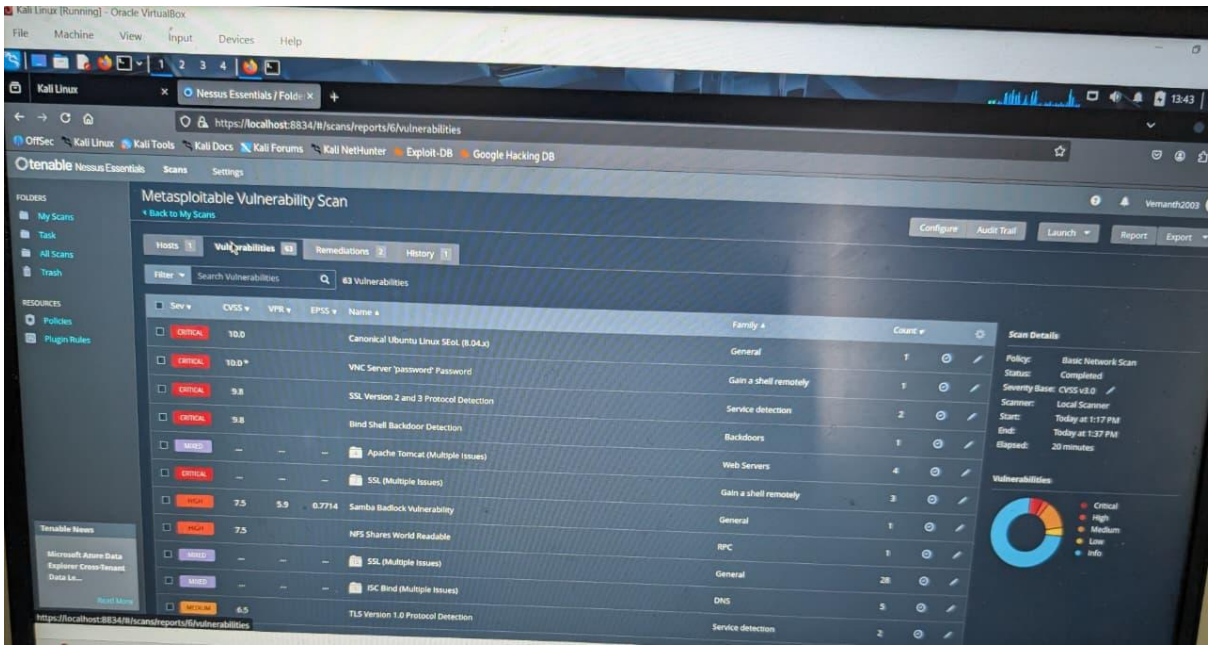

Figure: Detailed list of vulnerabilities detected by Nessus during the vulnerability scan.

The scan results identified multiple vulnerabilities across different severity levels, including critical and high-risk vulnerabilities. Examples of high severity findings included:

- Canonical Ubuntu Linux vulnerabilities
- VNC Server password vulnerability
- SSL Version 2 and 3 protocol detection
- Bind shell backdoor detection

These vulnerabilities have high CVSS scores, indicating that they could allow attackers to gain unauthorized access or exploit services running on the system.

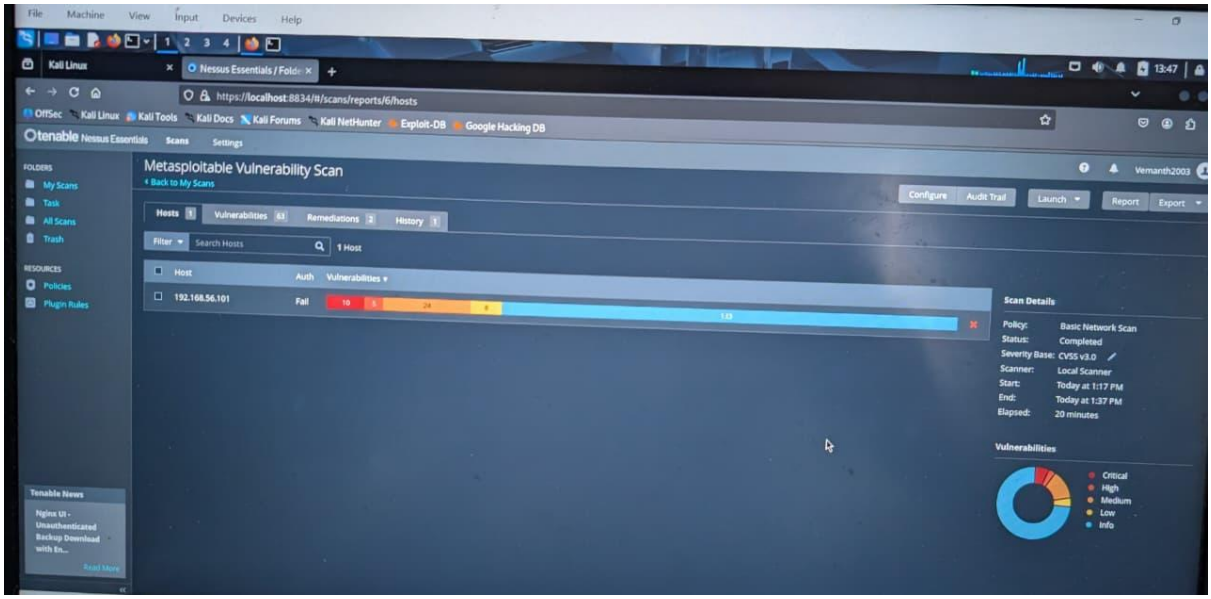

Figure: Host vulnerability summary for the Metasploitable2 system identified during the Nessus scan.

Nessus also provided remediation suggestions such as updating software versions, patching vulnerable services, and disabling insecure protocols.

4.3 Osquery Monitoring

Osquery is an open-source endpoint monitoring tool that allows users to query system information using SQL-like commands. It provides visibility into system processes, network connections, and operating system activity.

In this task, Osquery was installed on the Windows host system to monitor running processes. The interactive **osqueryi** shell was used to execute queries to retrieve information about system processes.

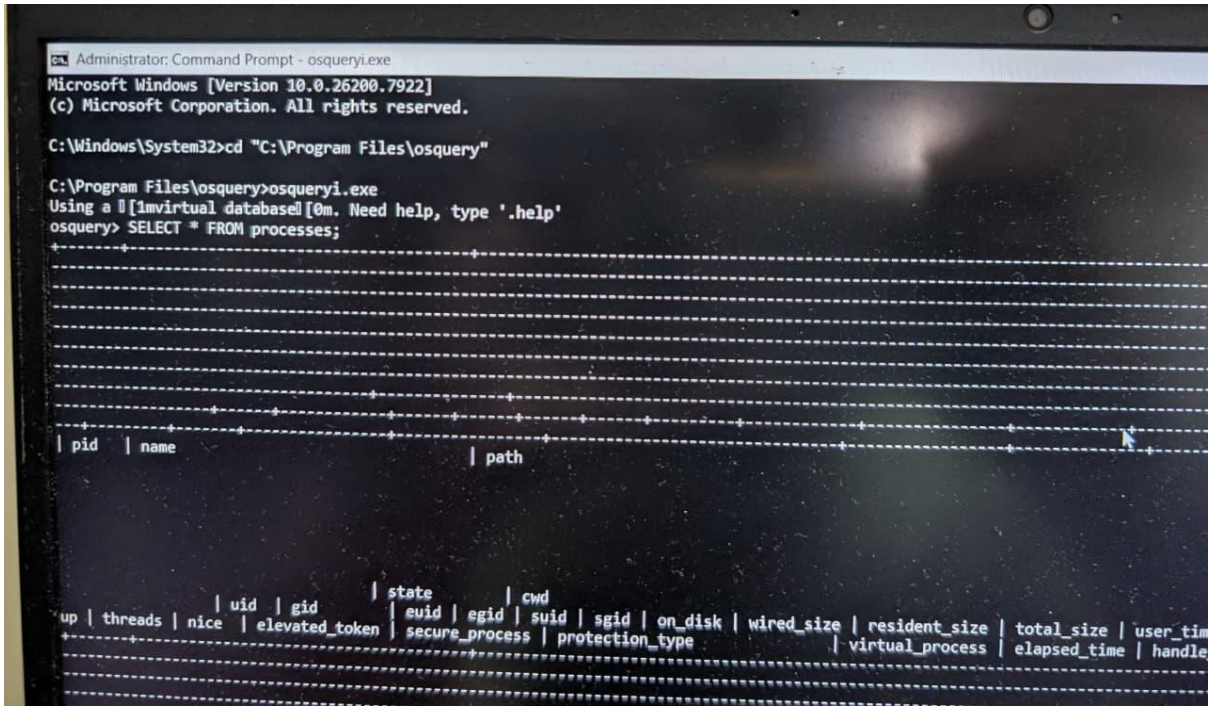

Figure: Running Osquery interactive shell on the Windows system to query process information

The following query was executed to list running command prompt processes:

```
SELECT name, pid, path FROM processes WHERE name='cmd.exe';
```

This query searches the system process table and displays all active **cmd.exe** processes.

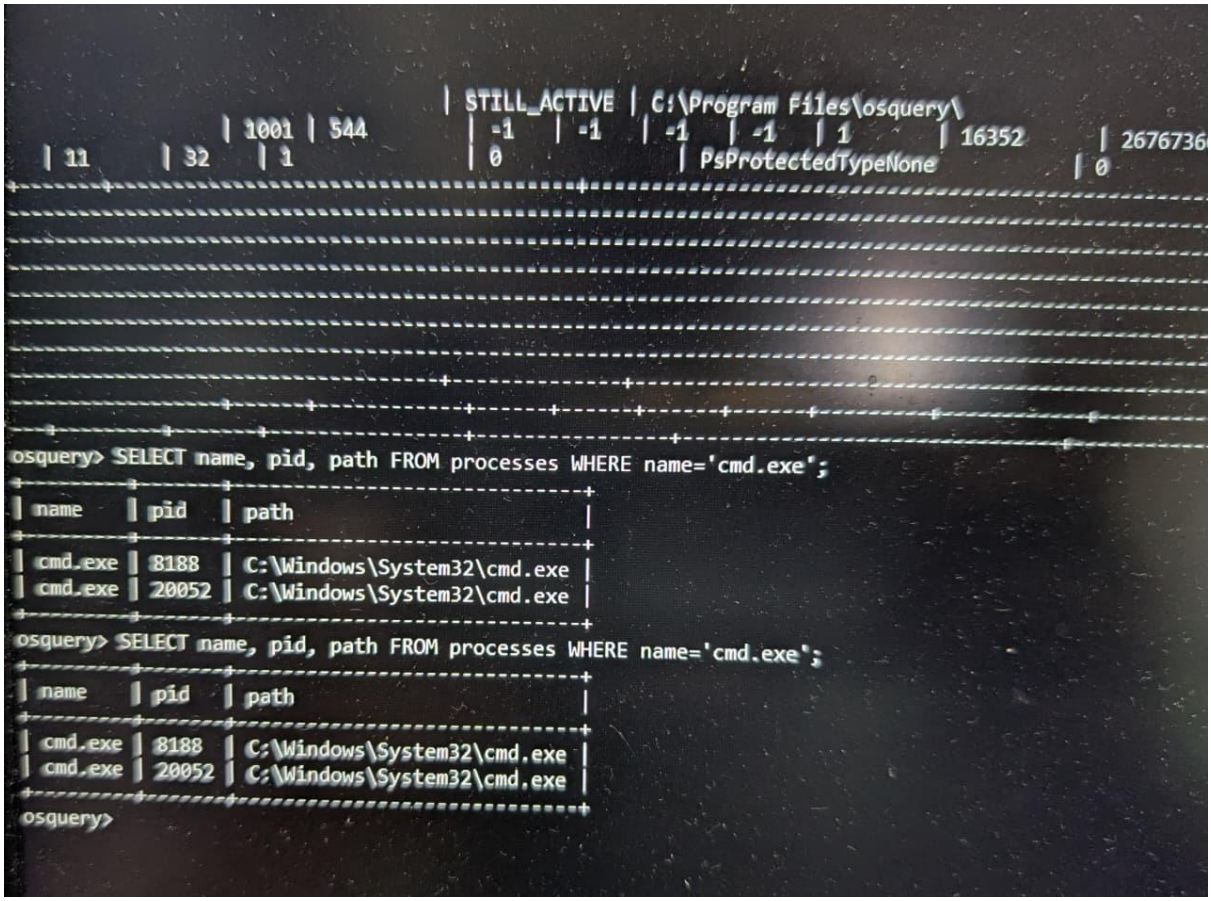

Figure: Osquery output displaying active cmd.exe processes with their process IDs and file paths

The results displayed active **cmd.exe** processes along with their process IDs and file paths, confirming that Osquery successfully detected the simulated activity.

This demonstrates how endpoint monitoring tools can be used to identify potentially suspicious processes running on a system and assist security analysts in monitoring endpoint behavior.

5. Basic Security Concepts

Understanding fundamental security concepts is essential for analyzing threats and protecting information systems. These concepts form the foundation of cybersecurity practices and help security professionals design systems that protect data and maintain operational stability.

One of the most important models in cybersecurity is the **CIA Triad**, which represents the three core principles of information security: **Confidentiality, Integrity, and Availability**.

- **Confidentiality** ensures that sensitive information is accessible only to authorized users. Techniques such as encryption, authentication mechanisms, and access control policies are used to maintain confidentiality.
- **Integrity** ensures that data remains accurate and unaltered unless modified by authorized individuals. Methods such as hashing, checksums, and digital signatures help maintain data integrity.
- **Availability** ensures that systems and data are accessible to authorized users whenever needed. Backup systems, redundancy, and disaster recovery mechanisms help maintain availability.

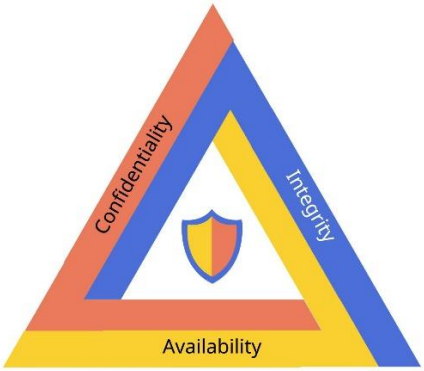

Figure: CIA Triad model representing the three core principles of information security.

Another important concept in cybersecurity is understanding the difference between **threats, vulnerabilities, and risks**.

- A **threat** is a potential cause of harm to a system, such as malware, hackers, or insider attacks.
- A **vulnerability** is a weakness in a system that can be exploited by a threat.
- A **risk** represents the potential damage that may occur when a threat exploits a vulnerability.

Security strategies often focus on reducing vulnerabilities and minimizing risks through proper system design and security controls.

Modern security architectures also emphasize **defense-in-depth** and **zero trust** principles.

Defense-in-depth involves implementing multiple layers of security controls throughout a system. Even if one layer fails, additional security mechanisms continue protecting the system.

Zero trust is a modern security model that assumes no user or device should be trusted by default. Every access request must be verified through authentication, authorization, and continuous monitoring.

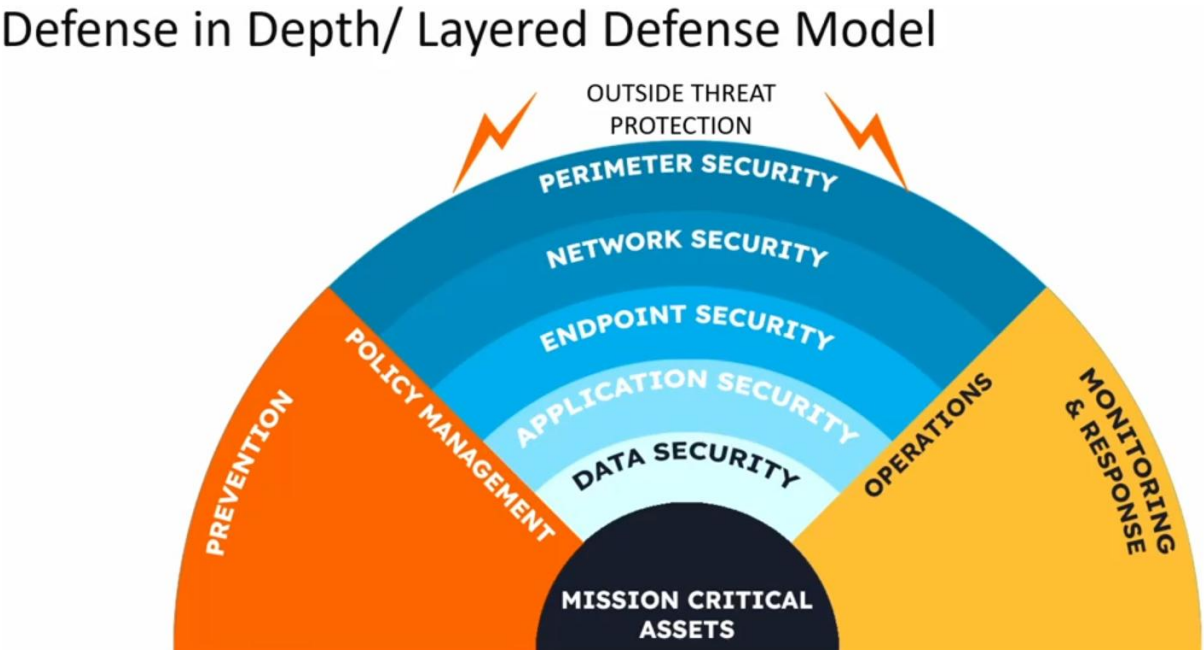

Figure: Defense-in-depth model illustrating multiple layers of security protection.

6. Security Operations Workflow

Security operations follow a structured workflow that helps organizations detect, analyze, and respond to security incidents effectively. Security teams continuously monitor systems and networks to identify suspicious activities and respond quickly to potential threats.

The workflow generally consists of four major stages: **Detection, Triage, Investigation, and Response**.

Detection

Detection is the process of identifying suspicious activities within a network or system. Security tools such as SIEM platforms, intrusion detection systems, and endpoint monitoring tools generate alerts when unusual behavior is detected. These alerts may indicate malware infections, unauthorized access attempts, or other potential threats.

Triage

Once an alert is generated, the security team performs triage to determine the severity and priority of the incident. Analysts evaluate alerts to determine whether they represent real threats or false positives. High-severity incidents are prioritized for immediate investigation.

Investigation

During the investigation phase, analysts gather and analyze relevant information related to the alert. This may include examining system logs, analyzing network traffic, and identifying indicators of compromise (IOCs). The goal is to determine the root cause of the incident and understand its impact.

Response

In the response phase, actions are taken to mitigate the threat. This may include isolating compromised systems, blocking malicious IP addresses, removing malware, or restoring systems from backups.

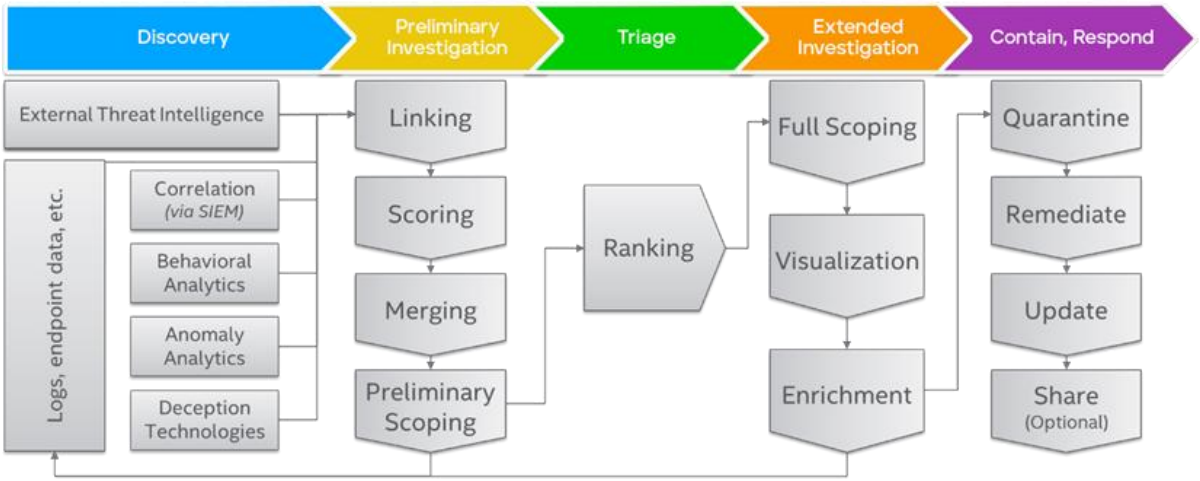

Figure: Security operations workflow showing detection, triage, investigation, and response stages.

Simulating security workflows using incident response platforms such as **TheHive** helps analysts practice incident management procedures. Creating incident flowcharts for scenarios such as phishing attacks helps security teams understand how threats are detected and handled in a structured manner.

7. Incident Response Basics

Incident response is the process of managing and responding to security incidents in order to minimize damage and restore normal system operations. Organizations follow structured frameworks to handle incidents effectively and consistently.

One widely used framework is defined in **NIST SP 800-61**, which describes the **Incident Response Lifecycle** consisting of six main phases.

Preparation

The preparation phase involves establishing security policies, training personnel, and implementing monitoring tools that enable organizations to detect and respond to incidents effectively.

Identification

During the identification phase, security teams determine whether an event represents a security incident. Analysts analyze alerts, logs, and system activity to confirm the presence of malicious activity.

Containment

Once an incident is confirmed, containment measures are implemented to prevent the attack from spreading. This may involve isolating compromised systems or blocking malicious network traffic.

Eradication

In the eradication phase, the root cause of the incident is removed from the system. Malware may be deleted, vulnerabilities patched, or compromised accounts reset.

Recovery

After the threat has been removed, systems are restored to normal operation. Monitoring continues to ensure the threat does not reoccur.

Lessons Learned

The final stage involves reviewing the incident and documenting what happened. This helps organizations improve their security procedures and strengthen defenses against future attacks.

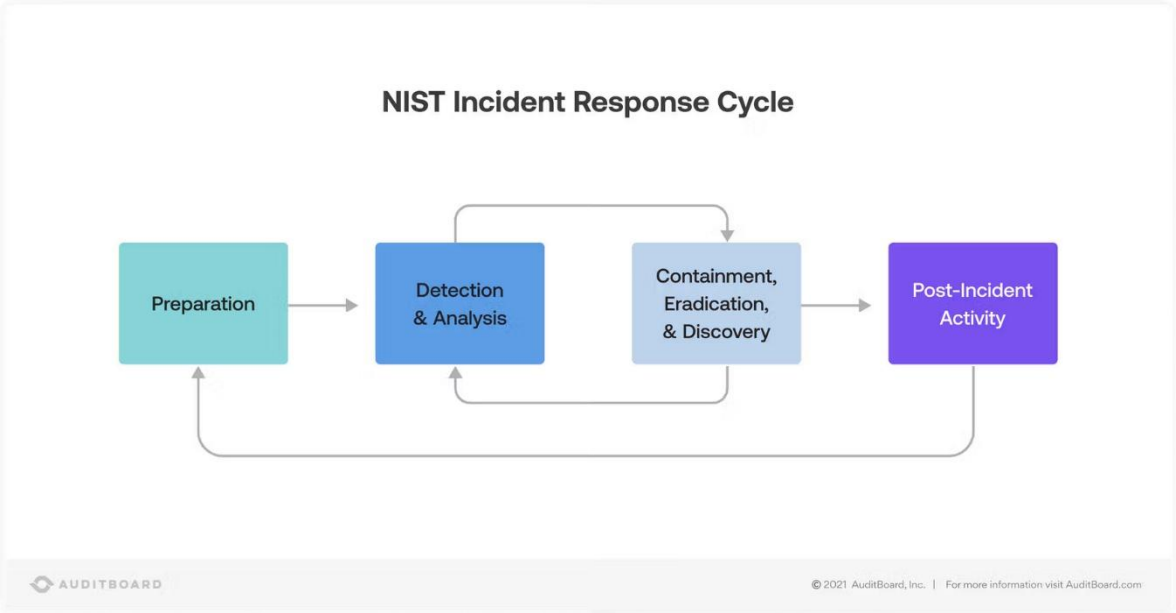

Figure: Incident response lifecycle based on the NIST SP 800-61 framework.

Organizations often conduct **tabletop exercises**, such as simulated ransomware attacks, to train security teams and test their incident response procedures.

8. Log Analysis Practice

Log analysis is an important activity in security monitoring because it allows analysts to detect suspicious activities, investigate incidents, and identify potential security threats. Security logs record system events such as login attempts, system errors, and configuration changes. By analyzing these logs, security analysts can identify abnormal behavior and respond to potential security incidents.

In this exercise, Windows Event Viewer was used to analyze system security logs and identify failed login attempts. The goal was to simulate a brute-force attack scenario and detect the failed authentication events recorded in the system logs.

8.1 Windows Event Viewer Log Analysis

Windows Event Viewer is a built-in tool used to monitor and analyze system logs in Windows operating systems. It records various events generated by the operating system and applications, including authentication events, system warnings, and security alerts.

In this task, the **Security log** in Windows Event Viewer was analyzed to detect failed login attempts. Failed authentication events are recorded using **Event ID 4625**, which indicates that a login attempt failed due to incorrect credentials.

To analyze these events, the Security log was filtered using the **Filter Current Log** option in Event Viewer. The filter was configured to display only events with **Event ID 4625**, allowing the identification of failed login attempts in the system.

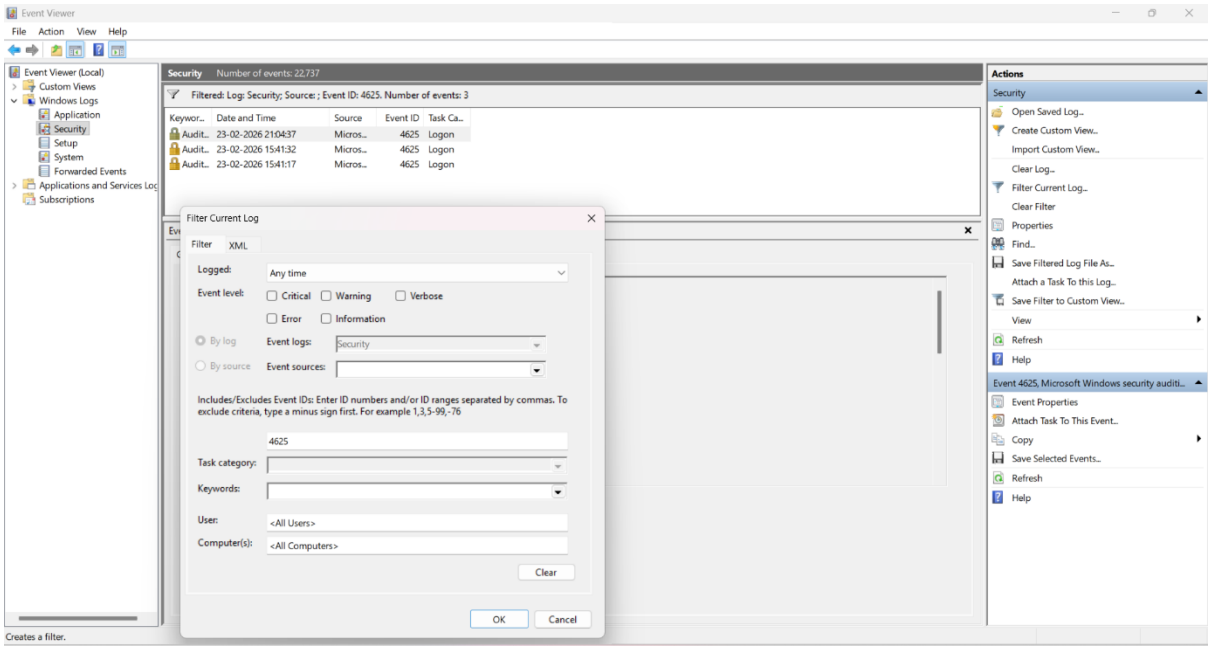

Figure: Windows Event Viewer filtered logs showing failed login attempts (Event ID 4625).

The filtered logs display important details such as the event ID, timestamp, user account involved, and the system that generated the event. These details help analysts identify suspicious login behavior and investigate potential security incidents.

8.2 Brute-Force Detection Simulation

To simulate a brute-force attack scenario, multiple incorrect login attempts were intentionally generated on the Windows system. A brute-force attack occurs when an attacker repeatedly attempts different passwords in order to gain unauthorized access to a system.

Several incorrect password attempts were performed on the Windows login screen. These failed authentication attempts were automatically recorded in the Windows Security logs as **Event ID 4625** entries.

After generating the failed login attempts, the Event Viewer logs were refreshed and multiple Event ID 4625 entries were observed. The presence of repeated failed login attempts within a short period of time can indicate a possible brute-force attack attempt.

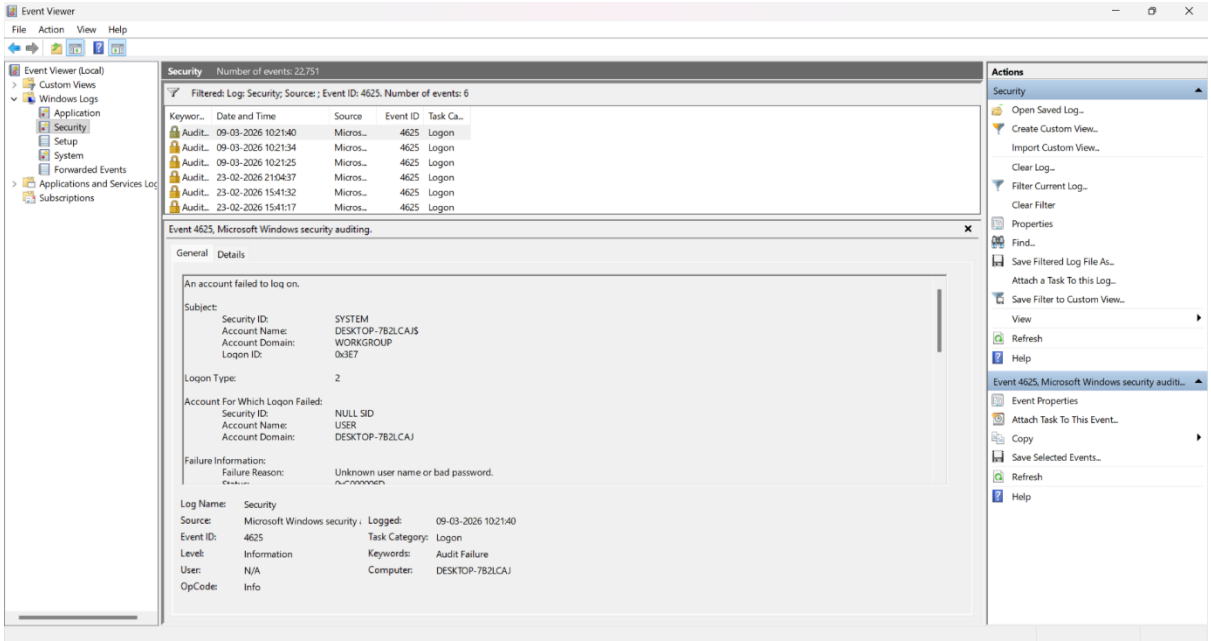

Figure: Multiple failed login attempts detected in Windows Security logs indicating a potential brute-force attack.

This exercise demonstrates how security analysts can monitor authentication logs to detect suspicious login activity.

8.3 Exporting Security Logs

Security logs can be exported for further analysis or documentation. Exporting logs allows analysts to store event records and analyze them using external tools or spreadsheet applications.

In this task, the filtered security logs containing Event ID 4625 were exported to a **CSV (Comma Separated Values)** file. This format allows the logs to be opened and analyzed using tools such as Microsoft Excel.

The export option was accessed by selecting **Save Filtered Log File As** in Event Viewer and choosing the CSV format.

Exporting logs is a common practice in security operations because it allows security teams to maintain records of security events and perform deeper analysis when necessary.

9. Document Security Events

Documenting security events is an important practice in cybersecurity operations. Security analysts record details of detected incidents in order to track suspicious activities, maintain investigation records, and improve incident response procedures. Proper documentation ensures that security teams have a clear understanding of what happened, when it occurred, and what actions were taken to address the issue.

In this task, a simple documentation template was created to record security events detected during log analysis. The template includes important information such as the date and time of the event, the source IP address, the event ID, a description of the event, and the actions taken by the analyst.

The following template was used to document security events:

Date/Time	Source IP	Event ID	Description	Action Taken
23-02-2026 21:04:37	Local Host, Microsoft Windows security auditing.	4625	Unknown user name or bad password	Investigated failed authentication attempt and reviewed Windows Security logs for suspicious activity
09-03-2026 10:21:40	Local Host, Microsoft Windows security auditing.	4625	Unknown user name or bad password	Monitored system for repeated login failures and confirmed no successful unauthorized access

This documentation example represents a scenario where multiple failed login attempts were detected in the Windows Event Viewer logs. The failed login events correspond to **Event ID 4625**, which indicates that authentication attempts were unsuccessful.

Recording these events allows security analysts to maintain an organized record of suspicious activities and track potential security incidents. If similar events occur repeatedly, analysts can investigate further to determine whether the activity represents a brute-force attack or another type of unauthorized access attempt.

Maintaining proper documentation is an essential part of security operations because it helps organizations maintain incident records, improve investigation processes, and support future security audits.

10.Set Up Monitoring Dashboards

Monitoring dashboards are essential in a Security Operations Center (SOC) as they allow analysts to visualize security events, identify suspicious patterns, and monitor system activity in real time. In this task, Kibana was used to create visualizations and dashboards based on log data collected through Logstash. The logs were accessed using the `logstash-*` data view in Kibana.

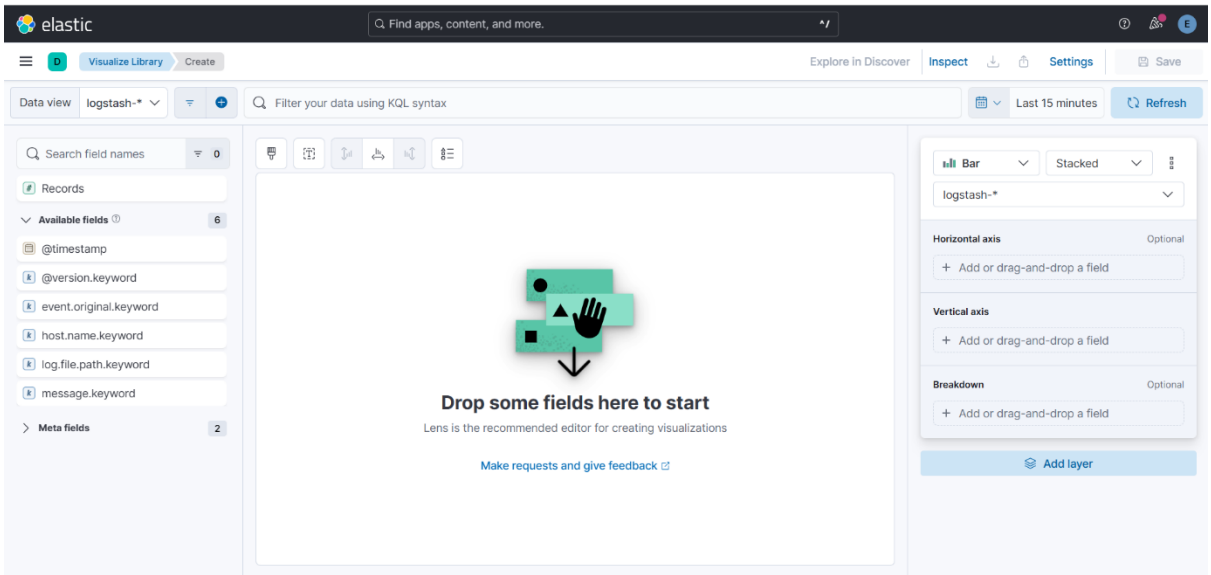

Figure: Creating in Visualize Library

First, a visualization was created to analyze the **frequency of events over time**. Using Kibana Lens, the `@timestamp` field was selected to generate a histogram chart that displays the number of log events occurring within specific time intervals. This visualization helps analysts understand patterns in log activity and identify unusual spikes that may indicate abnormal system behavior or potential security incidents.

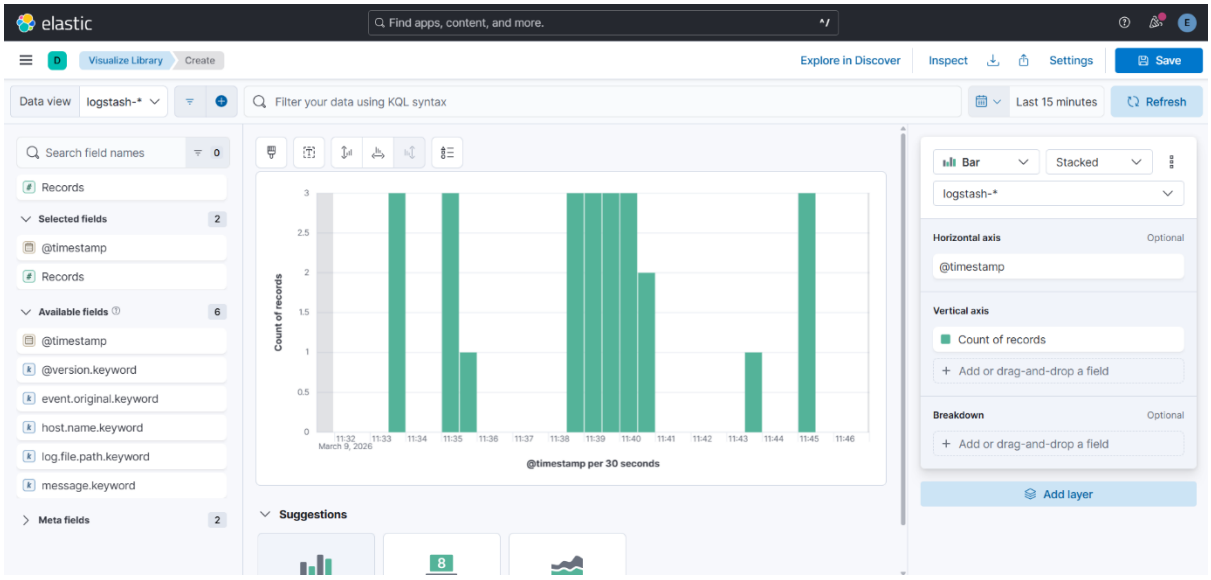

Figure: Histogram visualization in Kibana showing the frequency of security log events over time.

Next, another visualization was created to identify the **top sources generating log activity**. Since the available log data did not contain a `source.ip` field, the `host.name.keyword` field was used instead. A bar chart was generated to display the top hosts producing log events. This type of visualization allows analysts to quickly identify systems that generate a high volume of logs, which may indicate suspicious activity or potential security issues.

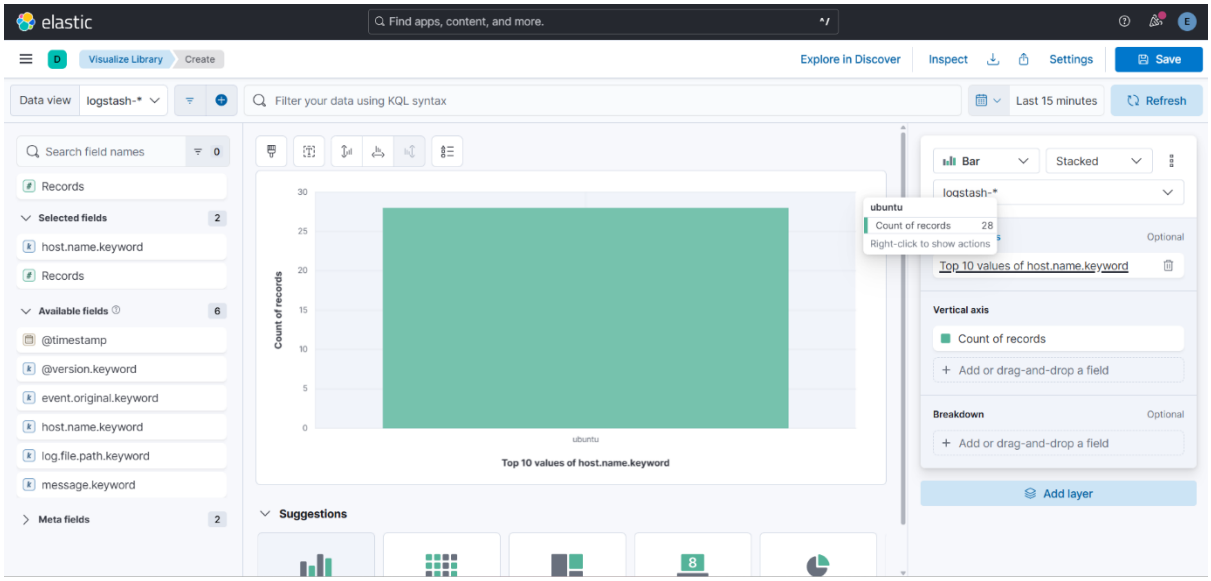

Figure: Bar chart visualization showing the top hosts generating log events in the system.

After creating the visualizations, a **monitoring dashboard** was developed in Kibana. The previously created charts were added to the dashboard to provide a centralized view of security activity. The dashboard allows analysts to monitor event frequency and identify the hosts responsible for generating logs. Such dashboards are widely used in SOC environments to support threat monitoring and incident detection by providing an overview of security-related events.

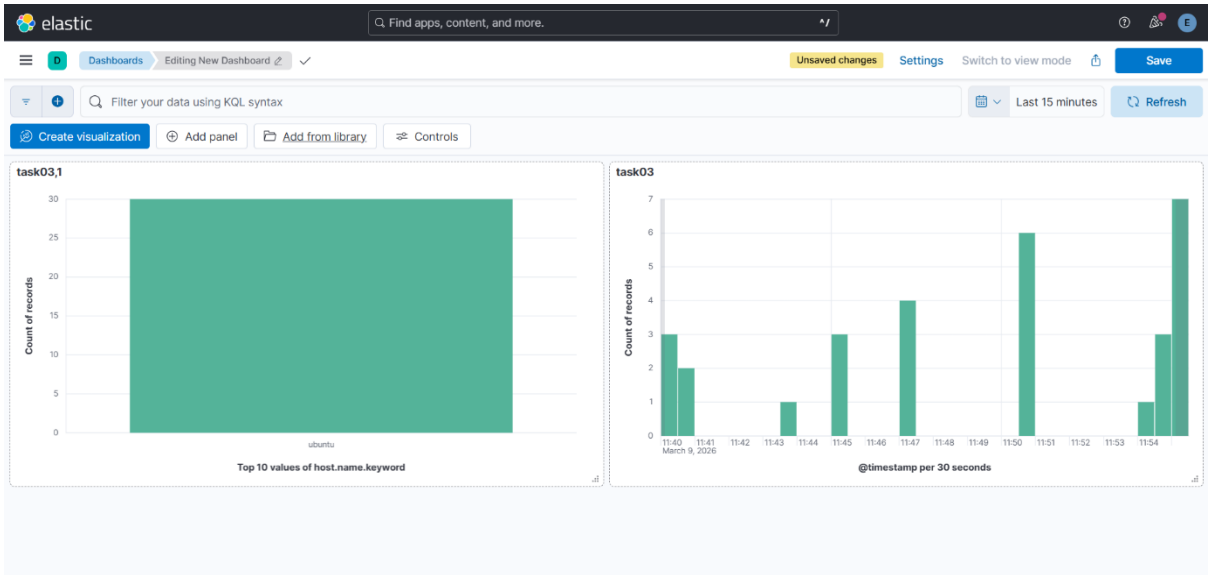

Figure: SOC monitoring dashboard displaying security events and top hosts generating logs.

Through this task, Kibana dashboards were successfully configured to visualize log data and support security monitoring. These visualizations help analysts detect abnormal patterns in system activity and improve situational awareness within a SOC environment.

NOTE:
I was able to complete the Kibana visualizations and dashboard using the data view. However, when I try to create an alert rule in **Security → Rules**, it shows **"Detection engine permissions required"**, and I cannot access the rule creation page.

Alerts

Detection engine permissions required

You do not have the required permissions for viewing the detection engine. For more help, contact your administrator.

View documentation